1 Integer ambiguity resolution

Integer ambiguity resolution (AR) converts the real-valued, bias-contaminated phase ambiguities of the float solution into the integer carrier-cycle counts they represent. The engine implements MLAMBDA [3] over a between-satellite-differenced ambiguity sub-problem extracted from the filter state and covariance. The operative quantity is the ambiguity differenced against a per-constellation pivot p_g (the differencing section of the Foundations document); the same integer-estimation machinery serves both the single-receiver and the baseline modes, but *what makes that differenced ambiguity an integer differs by mode*. The undifferenced ambiguity $N^i_{r,s}$ of the phase observation equation (Foundations, observation equations) carries a receiver phase bias $b_{r,s}$ and a satellite phase bias b^i_s. In single-receiver PPP / PPP-RTK the pivot difference is a *between-satellite single difference* $\Delta\,\cdot$ (the between-satellite single-difference section of the Foundations document), which cancels the receiver bias $b_{r,s}$ (common to every tracked satellite) but *not* the satellite bias; the residual $\Delta N^{i,p_g}_{r,s}$ is integer only once b^i_s has been removed by a consistent external phase-bias product (BIA / SSR, the SSR-corrections document). In baseline mode the state already holds true double differences $\nabla\Delta\,\cdot$ (the double-difference section of the Foundations document), where the additional between-receiver difference cancels the satellite bias as well, so $\nabla\Delta N$ is integer intrinsically. We retain the resolver's internal label $\nabla\Delta$ for the differenced ambiguity in the equations below (the algebra is identical either way), with the understanding that it denotes the between-satellite single difference Δ in the single-receiver case and the full double difference only in baseline mode. The implementation is split along a driver/kernel seam: `src/core/filter/AmbiguityResolver.cpp` is the *driver*, covering state-vector assembly, metre↔cycle rescaling, per-pair λ lookup, DD formation, the WL/NL decomposition, and application of the returned integer fix, while the decorrelation-agnostic *numerical core* (the LDL^{T} factorisation, the $\boldsymbol{Z}$-reduction, the Schnorr–Euchner search, the bootstrap success rate, and the full acceptance battery, all as pure free functions over ($\hat{\boldsymbol{N}}$ cycles, $\boldsymbol{Q}_{\hat{N}}$ cycles^2) with no `StateManager` / `KalmanFilter` / config coupling) lives in the `GnssProcessor::ils` namespace of `src/core/filter/IlsCore.{hpp,cpp}`. That same `ils` core is the single implementation the PPP-RTK network producer also calls (the Network mode document), though there it runs under its own gate set and purely as a read-only fixability probe, since the producer is float-only at the time of writing (Section 1.7). Configuration for the single-receiver and baseline paths described below is under the JSON path `estimation.ambiguity_resolution.*`.

1.1 The integer least-squares problem

The float solution delivers a real-valued ambiguity estimate $\hat{\boldsymbol{N}}$ (in cycles) with covariance $\boldsymbol{Q}_{\hat{N}}$, both obtained by sub-vector extraction from the filter state $\boldsymbol{x}$ and covariance $\boldsymbol{P}$. Integer ambiguity resolution is the weighted closest-lattice-point problem

$$\boxed{\check{\boldsymbol{N}} \;=\; \arg\min_{\boldsymbol{a}\,\in\,\mathbb{Z}^n} \big(\hat{\boldsymbol{N}}-\boldsymbol{a}\big)^{\mathrm{T}}\,\boldsymbol{Q}_{\hat{N}}^{-1}\,\big(\hat{\boldsymbol{N}}-\boldsymbol{a}\big),} \tag{1}$$

i.e. minimisation of the squared Mahalanobis distance from $\hat{\boldsymbol{N}}$ to a lattice point $\boldsymbol{a}$ in the metric of $\boldsymbol{Q}_{\hat{N}}^{-1}$. This is an NP-hard integer least-squares (ILS) program; LAMBDA and its MLAMBDA variant make it tractable in three invariant stages: (i) an LDL^{T} factorisation of $\boldsymbol{Q}_{\hat{N}}$, (ii) a decorrelating integer $\boldsymbol{Z}$-transform (lattice reduction), and (iii) a depth-first shrinking-ellipsoid

search over the reduced lattice, none of which carries a tuning parameter. We follow the hat/check convention of the notation block: $\hat{\boldsymbol{N}}$ is the float estimate, $\check{\boldsymbol{N}}$ the integer-fixed solution.

Float extraction and rescaling to cycles. The filter state stores per-band ambiguity states in *metres* ($\lambda_s N$, one per tracked (satellite, signal)), not cycles, so the resolver first forms the between-satellite-differenced float and covariance by explicit differencing against the pivot p_g and then rescales by wavelength. (In baseline mode the state already holds the double difference and the differencing step below collapses to a direct selection; see Section 1.6.) For a differenced pair k with satellite index $i(k)$ and pivot index $p(k)$ (`src/core/filter/AmbiguityResolver.cpp`):

$$y_k^{\nabla\Delta} = \boldsymbol{x}_{i(k)} - \boldsymbol{x}_{p(k)}, \qquad \text{(DD float, m)} \tag{2}$$

$$\boldsymbol{Q}_{kl}^{\nabla\Delta} = \boldsymbol{P}_{i_k i_l} - \boldsymbol{P}_{i_k p_l} - \boldsymbol{P}_{p_k i_l} + \boldsymbol{P}_{p_k p_l}, \qquad \text{(DD covariance, m}^2\text{)} \tag{3}$$

$$\hat{\boldsymbol{N}}_k = y_k^{\nabla\Delta}/\lambda_k, \qquad (\boldsymbol{Q}_{\hat{N}})_{kl} = \boldsymbol{Q}_{kl}^{\nabla\Delta}/(\lambda_k\lambda_l). \qquad \text{(cycle domain)} \tag{4}$$

The rescaling (4) is load-bearing: running MLAMBDA on metre-domain floats would round toward integer-*metre* values with no carrier-cycle semantics, and the ratio test (Section 1.5) would degenerate to ≈ 1 irrespective of float quality. The per-pair wavelength λ_k is resolved with GLONASS FDMA per-satellite frequencies where applicable, and $\boldsymbol{Q}^{\nabla\Delta}$ is symmetrised, $\boldsymbol{Q}^{\nabla\Delta} \leftarrow \frac{1}{2}(\boldsymbol{Q}^{\nabla\Delta} + \boldsymbol{Q}^{\nabla\Delta\mathrm{T}})$, before factorisation. The resolver elects its own DD reference: the highest-elevation member of each (constellation, signal) group among the AR-eligible ambiguities, with all remaining members of the group differenced against it. This is deliberately *not* the filter's BSSD pivot: under BSSD the pivot's ambiguity is frozen at exactly zero variance, which excludes it from the candidate set (Section 1.4), so the resolver differences against a second reference and the resulting quantity is a difference of single differences. The two elections also run on different rules: the BSSD pivot is chosen on a hard phase-coverage tier plus a convergence/elevation score and an ephemeris-freshness bar (the differencing section of the Foundations document), the AR reference on elevation alone (`AmbiguityResolver::buildDdPairs`).

1.2 Decorrelation: LDL^{T} factorisation and the integer Z-transform

Factorisation. MLAMBDA factorises the symmetric positive-definite ambiguity covariance as

$$\boldsymbol{Q}_{\hat{N}} \;=\; \boldsymbol{L}^{\mathrm{T}}\,\boldsymbol{D}\,\boldsymbol{L}, \qquad \boldsymbol{L} \text{ unit lower-triangular}, \;\; \boldsymbol{D} = \mathrm{diag}(d_1, \ldots, d_n), \;\; d_i > 0, \tag{5}$$

factored in reverse index order (the Chang–Yang MLAMBDA convention). A non-positive pivot $d_i \le 0$ aborts the fix, since it signals a non-PD covariance. After reduction, d_i is the *conditional variance* of the i-th decorrelated ambiguity given the later ones; it feeds the success-rate gate of Section 1.5 directly.

The Z-transform. The central device of LAMBDA [3] is a unimodular integer transform $\boldsymbol{Z} \in \mathbb{Z}^{n\times n}$, $|\det \boldsymbol{Z}| = 1$, that re-parameterises

$$\boldsymbol{z} = \boldsymbol{Z}^{\mathrm{T}}\hat{\boldsymbol{N}}, \qquad \boldsymbol{Q}_{\hat{z}} = \boldsymbol{Z}^{\mathrm{T}}\,\boldsymbol{Q}_{\hat{N}}\,\boldsymbol{Z}, \tag{6}$$

so that the transformed ambiguities are nearly decorrelated (small off-diagonal $\boldsymbol{L}$ entries) and their conditional variances are ordered. We write the decorrelated float vector as $\bar{\boldsymbol{z}} := \boldsymbol{Z}^{\mathrm{T}}\hat{\boldsymbol{N}}$ (the quantity the search of Section 1.2 actually rounds; implemented as `z_float = Z.transpose()*a` in `ils::mlambda` (`src/core/filter/IlsCore.cpp`)), with components $\bar{z}_k$. Because $\boldsymbol{Z}$ is unimodular

the integer minimiser maps back exactly, $\check{\boldsymbol{N}} = (\boldsymbol{Z}^{\mathrm{T}})^{-1}\check{\boldsymbol{z}}$ with $(\boldsymbol{Z}^{\mathrm{T}})^{-1}$ also integer; decorrelation therefore does not alter the solution of (1); it only makes the search ellipsoid closer to spherical, collapsing the enumeration count by orders of magnitude. The reduction interleaves two operations sweeping $j = n - 2$ downward.

Integer Gauss transformations zero the magnitude of sub-diagonal $\boldsymbol{L}$ entries by integer column operations,

$$\mu = \lceil L_{ij} \rfloor, \qquad \boldsymbol{L}_{(:,j)} \mathrel{-}= \mu\, \boldsymbol{L}_{(:,i)}, \qquad \boldsymbol{Z}_{(:,j)} \mathrel{-}= \mu\, \boldsymbol{Z}_{(:,i)}, \tag{7}$$

where $\lceil \cdot \rfloor$ is round-to-nearest, keeping every $|L_{ij}| \le \frac{1}{2}$.

Permutations (conditional adjacent swaps) reorder ambiguities when doing so reduces the later conditional variance. With $\bar{d}_j = d_j + L_{j+1,j}^2\, d_{j+1}$, the swap fires when

$$\bar{d}_j + \varepsilon \;<\; d_{j+1}, \qquad \varepsilon = 10^{-6}, \tag{8}$$

after which d_j, d_{j+1}, the affected rows/columns of $\boldsymbol{L}$ and the columns of $\boldsymbol{Z}$ are updated by the standard rank-one formulas ($\eta = d_j/\bar{d}_j$, $\lambda = d_{j+1}L_{j+1,j}/\bar{d}_j$, $d_j \leftarrow \eta\, d_{j+1}$, $d_{j+1} \leftarrow \bar{d}_j$), and the sweep restarts from $j = n - 2$. The guard $\varepsilon = 10^{-6}$ prevents thrashing on near-equal variances and is a numerical safeguard, not an estimator knob; it has no config exposure.

The shrinking-ellipsoid search. The reduced problem is solved by depth-first enumeration of lattice points inside the ellipsoid $\sum_i (z_i - \hat{z}_i)^2/d_i \le \chi^2$, in Schnorr–Euchner order (nearest integer first, then zig-zagging outward). Recursing from the last level $k = n - 1$ down to 0,

$$\hat{z}_k = \bar{z}_k + \sum_{i>k} (z_i - \hat{z}_i)\, L_{ik}, \qquad \text{(conditional centre)} \tag{9}$$

$$z_k = \lceil \hat{z}_k \rfloor, \qquad \mathrm{dist}_k = \mathrm{dist}_{k+1} + \frac{(\hat{z}_k - z_k)^2}{d_k}. \qquad \text{(partial distance)} \tag{10}$$

When a complete integer vector reaches level 0 with cumulative distance below the current bound χ^2, it is recorded and the bound shrinks to the worst retained candidate, contracting the ellipsoid as better candidates are found. The search (`ils::search` in `src/core/filter/IlsCore.cpp`) retains the $m = 2$ best candidates (for the ratio test) sorted by ascending squared distance, under a hard iteration cap (`kMaxIter` = 10 000). Its outputs are the two smallest objective values $s_1 \le s_2$ of (1) and the best integer vector $\check{\boldsymbol{z}}$, back-transformed via $\check{\boldsymbol{N}} = (\boldsymbol{Z}^{\mathrm{T}})^{-1}\check{\boldsymbol{z}}$.

1.3 Wide-lane / narrow-lane decomposition (PPP path)

Standard PPP and PPP-RTK AR use a *wide-lane-first* decomposition when Melbourne–Wübbena (MW)-resolved wide-lane (WL) integers are available, halving the search dimension and exploiting the long WL wavelength [2]. The driver attempts the WL-constrained narrow-lane (NL) fix before any unconstrained LAMBDA and, critically, does *not* fall through to an unconstrained search when WL information exists but the constrained fix fails: a wrong-WL fix is worse than no fix, so "no NL AR for satellites whose WL is float" is the enforced hierarchy (`src/core/filter/AmbiguityResolver.cpp`).

WL from Melbourne–Wübbena (integer by contract). The DD wide-lane integer is differenced directly from the MW detector's single-difference WL integers (`src/core/slip/MwSlipDetector`), not searched:

$$\nabla\Delta N_{\mathrm{WL}} = N_{\mathrm{WL}}^{i} - N_{\mathrm{WL}}^{p_g} \in \mathbb{Z}, \qquad \lambda_{\mathrm{WL}} = \frac{c}{f_1 - f_2}. \tag{11}$$

NL search variable. For each dual-frequency DD pair the float narrow-lane ambiguity (ionosphere-free phase combination, in NL cycles) is formed by the transform $\boldsymbol{T}$,

$$\hat{n}_{\mathrm{NL}} = \frac{f_1\,\nabla\Delta N_1 + f_2\,\nabla\Delta N_2}{f_1+f_2} = \frac{1}{f_1+f_2}\left(\frac{f_1}{\lambda_1}\,\nabla\Delta_1 + \frac{f_2}{\lambda_2}\,\nabla\Delta_2\right), \qquad \lambda_{\mathrm{NL}} = \frac{c}{f_1+f_2}, \tag{12}$$

with reduced covariance $\boldsymbol{Q}_{\mathrm{NL}} = \boldsymbol{T}\,\boldsymbol{Q}^{\nabla\Delta}\,\boldsymbol{T}^{\mathrm{T}}$. Only NL-paired DDs enter LAMBDA; unpaired single-signal DDs are excluded so they cannot contaminate the ratio test.

Integer-recentring shift. The float $\hat{n}_{\mathrm{NL}}$ is *not* integer-valued in general: substituting $N_2 = N_1 - \mathrm{WL}$ gives $\hat{n}_{\mathrm{NL}} = N_1 - \frac{f_2}{f_1+f_2}\,\mathrm{WL}$, which for GPS L1/L2 with $\mathrm{WL} = 1$ sits at $N_1 - 0.4379$, so neither integer is meaningfully closer and the ratio test degenerates near 1.6 regardless of float quality. The resolver therefore adds the known-WL offset so the search variable is integer in expectation,

$$\tilde{n}_k = \hat{n}_{\mathrm{NL},k} + \frac{f_2}{f_1+f_2}\,\nabla\Delta N_{\mathrm{WL},k} \;\approx\; N_{1,k} \;\in\; \mathbb{Z} + \text{noise}, \tag{13}$$

runs MLAMBDA on $\tilde{\boldsymbol{n}}$ (a constant per-row shift leaves $\boldsymbol{Q}_{\mathrm{NL}}$ unchanged), then back-shifts $\check{n}_{\mathrm{NL}} = \check{N}_1 - \frac{f_2}{f_1+f_2}\,\mathrm{WL}$.

Back-transform to per-signal DDs. The fixed integers map back to the two original signal DDs in metres,

$$\nabla\Delta_1^{\mathrm{fix}} = \left(\check{n}_{\mathrm{NL}} + \tfrac{f_2}{f_1+f_2}\,\nabla\Delta N_{\mathrm{WL}}\right)\lambda_1, \qquad \nabla\Delta_2^{\mathrm{fix}} = \left(\check{n}_{\mathrm{NL}} - \tfrac{f_1}{f_1+f_2}\,\nabla\Delta N_{\mathrm{WL}}\right)\lambda_2, \tag{14}$$

which re-enter the filter through the constrained update of Section 1.6.

State-basis variants. When `estimation.state_ambiguity_basis = nl_wl` the state holds NL/WL ambiguities directly and there are no per-band DD pairs to enumerate; today only the MW-resolved WL integers are applied as constraint rows (a full NL LAMBDA search with WL pre-constrained is future work in this basis). The complementary WL-only mode (`estimation.ambiguity_resolution.wl_only`, in the `per_band` basis) applies the cycle-domain constraint $\nabla\Delta_1/\lambda_1 - \nabla\Delta_2/\lambda_2 = \nabla\Delta N_{\mathrm{WL}}$ as a single pseudo-observation per pair without any NL search, with covariance inflation on the affected single-difference states when MW re-locks at a new integer so the state snaps within one epoch. Both WL-constraint paths (this one and the `nl_wl`-basis constraint below) build their pseudo-observation noise from a *separate* knob, $\boldsymbol{R} = \sigma_{\mathrm{wl}}^2\,\boldsymbol{I}$ with $\sigma_{\mathrm{wl}}^2 =$ `wl_only_pseudo_var` (10^{-4} cycles^2, note the cycle domain), *not* the metre-domain σ_{hold}^2 of the NL/DD feedback in (19). This `per_band wl_only` path carries no cross-frequency consistency check: it gates each DD WL only on the per-arc fractional residual (`wl_max_frac`, default 0.40, both arcs). The cross-frequency parity check lives instead in the `nl_wl` basis (`applyMwWlConstraintsNlWl_` in `src/core/filter/AmbiguityResolver.cpp`), where it rejects all three WL DDs of a triple-frequency satellite when the telescoping canonical cycle-count consistency $\nabla\Delta_{ab} - \nabla\Delta_{ac} + \nabla\Delta_{bc} \neq 0$ fails; it does *not* run in the `per_band wl_only` path.

1.4 Partial / subset ambiguity resolution

Partial AR operates at *two* levels: a static upstream candidate gate that screens which ambiguities ever enter the integer problem, and an optional dynamic backward-elimination search that recovers a fixable subset when the full set fails validation.

Upstream candidate gating (always on). Subset selection begins when the DD pairs are built (`src/core/filter/AmbiguityResolver.cpp`). A satellite/signal ambiguity is admitted to the integer set only if it passes every filter:

- **Initialised**: the state entry is non-zero (seeded by data), and **non-frozen**: its variance is non-zero (excludes a BSSD-frozen pivot).
- **Elevation**: $\theta_r^i \geq \theta_{\mathrm{AR}}$ (`elevation_mask`).
- **Resolvable**: the satellite is not flagged unresolvable for want of bias/ATX (BIA, ATX) coverage (single-receiver path; in baseline DD the analogous filter is the `exclude_keys` set of non-integer-capable DDs from `MeasurementModel`).
- **Constellation**: $g(i) \in$ `active_systems`, when that list is non-empty.
- **Pairable**: the (constellation, signal) group holds ≥ 2 satellites, so a pivot difference exists.
- **Mature** (`min_arc_info`, default 0 = off): both constituent arcs of a candidate have accumulated at least $\sum_k \sin^2 \theta_{r,k}^i$ of elevation-weighted code information since arc birth. Maturity is measured in *information*, not epochs, because an arc climbing off the mask and one at zenith accumulate at very different rates under the elevation stochastic model, so two arcs of equal age are not equally determined. The counter restarts on *both* ways an arc can end: the key leaving the state vector, and a refloat in place, which keeps the key and would otherwise let a fresh arc inherit its pre-slip age (`AmbiguityResolver::refreshArcAges_`, keyed on `StateManager::isAmbiguityPendingReseed`). Scope is narrow: the gate is tested only on the WL-constrained NL candidate set of Section 1.3, not on the baseline DD, WL-only, or unconstrained full-set paths. Its purpose is defensive, since one immature candidate spoils the joint ratio test of (15) for the whole set, but it is a wrong-fix guard, not a convergence accelerator: it admits a candidate once it is already well determined, which is precisely when the float needs least help.

The constellation filter keeps a system in the *float* solution while excluding it from the candidate set, useful when one constellation's residual model bias would drag down the ratio test.

Dynamic backward-elimination subset (`enable_partial_ar`, default off). With the candidate set fixed, the default behaviour is all-or-nothing: the full set is searched and either passes all of Section 1.5 or the epoch stays float. When `enable_partial_ar` is set, a failure of the full-set battery triggers a greedy backward elimination (`ils::selectFixableSubset` in `src/core/filter/IlsCore.cpp`, reached through the driver-side `selectFixableSubset_` config adapter) instead of falling straight back to float. Starting from the full candidate set, each iteration re-runs the complete MLAMBDA pipeline of Sections 1.1 to 1.2 on the current subset and re-tests the *same* acceptance battery: ratio lower/upper bound (15), bootstrap success rate (16), and per-entry fractional cap (17). On any failure it drops one entry and retries, down to a single ambiguity:

- if the *fractional cap* is the failing gate, the entry with the largest $|\hat{\boldsymbol{N}}_k - \check{\boldsymbol{N}}_k|$ (nearest a half-cycle boundary) is removed;
- otherwise the entry with the largest formal float variance $(\boldsymbol{Q}_{\hat{N}})_{kk}$, the least precise ambiguity, whose removal most improves both the bootstrap rate and the search conditioning, is removed.

Because the accepted subset is always re-validated by the full battery on its own covariance sub-block, this can only fix *fewer* ambiguities than the full set, never a worse-conditioned set: a wrong elimination order costs fixes, not integrity, and the path is bit-identical to the legacy behaviour when off. The same routine serves the per-band SD, baseline DD, and WL-constrained NL paths uniformly, operating purely in the decorrelation-agnostic cycle domain; each caller maps the surviving indices back to its own state rows and feeds them through the constrained update of Section 1.6.

1.5 Validation: ratio test and fixed-failure-rate gate

A fix is committed only if it passes all three of the following gates (a logical AND). The ratio and fixed-failure-rate (FFR) gates are theoretically distinct: one bounds the discrimination between the two best candidates, the other bounds the absolute success probability.

Ratio test (discrimination).

$$\boxed{\text{ratio} = \frac{s_2}{s_1} \;\geq\; \tau_{\text{ratio}}, \qquad s_1 \leq s_2 \text{ the two smallest ILS objectives,}} \tag{15}$$

with s_1, s_2 from the search of Section 1.2 and threshold τ_{ratio} (`ratio_threshold`). A second upper-bound rejection ratio $> \tau_{\text{ratio}}^{\max}$ (`ratio_upper_bound`) catches the pathological case of a degenerate covariance direction, a near-zero s_1 left by a previous wrong-integer hold, that would otherwise produce ultra-confident but meaningless ratios.

Fixed-failure-rate gate (bootstrap success rate). A geometry-driven, self-tuning lower bound on the ILS success probability,

$$\boxed{P_{s,\text{boot}} = \prod_{i=1}^{n} \operatorname{erf}\!\left(\frac{1}{2\sqrt{2\,d_i}}\right) \;\geq\; P_s^{\min},} \tag{16}$$

where the d_i are the post-reduction LDL^{T} conditional variances of (5) and $P_s^{\min}$ is `min_success_rate`. Per dimension the factor is $\Pr\big(|\mathcal{N}(0, d_i)| < \frac{1}{2}\big)$, the probability that sequential bootstrapped rounding lands on the correct integer, and the product is the standard Teunissen lower bound on the true ILS success rate. This ties the acceptance threshold to the actual problem geometry rather than to a fixed ratio alone.

Per-pair residual backstop. The joint ratio test can pass on a borderline set in which one pair sits near a half-cycle boundary; the fix is rejected if any single DD's fractional residual exceeds a cap,

$$\max_k \big|\hat{\boldsymbol{N}}_k - \check{\boldsymbol{N}}_k\big| \;\leq\; \kappa_{\text{frac}} \quad \text{(cycles)}, \tag{17}$$

with $\kappa_{\text{frac}} =$ `nl_max_per_pair_frac`.

Convergence guards. Before any search the resolver short-circuits AR while the float is immature, meaning the epoch count is below `min_epochs` or the position standard deviation $\sqrt{\boldsymbol{P}_{00} + \boldsymbol{P}_{11} + \boldsymbol{P}_{22}}$ exceeds `max_position_sigma_m`, because a marginal fix accepted before the covariance settles can lock onto wrong integers and then poison every subsequent epoch.

1.6 Feedback: constrained pseudo-observation update

Accepted integers are *not* fed back by the classical perfect back-substitution (the $R = 0$ conditioning form)

$$\boldsymbol{x}_{\text{fix}} = \boldsymbol{x} - \boldsymbol{Q}_{\boldsymbol{aa}}\, \boldsymbol{Q}^{\nabla\Delta-1}\big(y^{\nabla\Delta} - \check{\boldsymbol{N}}\big), \qquad \boldsymbol{P}_{\text{fix}} = \boldsymbol{P} - \boldsymbol{Q}_{\boldsymbol{aa}}\, \boldsymbol{Q}^{\nabla\Delta-1}\boldsymbol{Q}_{\boldsymbol{aa}}^{\mathrm{T}}, \tag{18}$$

but via a *finite-variance Joseph-form pseudo-observation* (`src/core/filter/AmbiguityResolver.cpp`; kernel `KalmanFilter::josephPseudoUpdate` in `include/filter/KalmanFilter.hpp`). The reason is structural: (18) drives the fixed ambiguity rows of $\boldsymbol{P}$ to ≈ 0 variance, so the next

epoch's LAMBDA sees a near-singular $\boldsymbol{Q}^{\nabla\Delta}$ and emits meaningless inflated ratios that trip the upper-bound guard of (15) indefinitely. A finite-variance pseudo-observation is *intended* to keep $\boldsymbol{P}$ well-conditioned so that subsequent epochs can re-confirm the integer (asymptoting toward it) or detect a slip through its residual. Whether it in fact does so is an open question at the time of writing: the resolver carries a read-only conditioning probe (`qbCondition` / `rebuildQbCycles` in `src/core/filter/AmbiguityResolver.cpp`) that eigen-decomposes the cycle-domain DD covariance either side of every fix and reject, added precisely because tightening σ^2_{hold} was observed to *reduce* the fix count and destabilise the kinematic solution, the signature of a covariance being driven singular by the constraint rather than merely tightened by it. The mechanism argued for below is therefore the design intent, not a validated result.

For each accepted DD pair k the constraint row, innovation and noise are

$$\boxed{\boldsymbol{H}_{(k,\,i)} = +1, \quad \boldsymbol{H}_{(k,\,p)} = -1, \qquad v_k = \lambda_k\,\check{\boldsymbol{N}}_k - \big(\boldsymbol{x}_i - \boldsymbol{x}_p\big), \qquad \boldsymbol{R} = \sigma^2_{\text{hold}}\,\boldsymbol{I},} \tag{19}$$

followed by $(\boldsymbol{x}_{\text{fix}}, \boldsymbol{P}_{\text{fix}}) = \textsc{JosephUpdate}(\boldsymbol{x}, \boldsymbol{P}, \boldsymbol{H}, \boldsymbol{v}, \boldsymbol{R})$, with $\sigma^2_{\text{hold}} =$ `hold_variance`. The DD/baseline path uses a single-column selector $\boldsymbol{H}_{(k,\,\text{idx})} = +1$ since the corresponding state value already *is* the double difference. Fix-and-hold mode (`estimation.ambiguity_resolution.mode` = 3) re-applies the same constraint each epoch to suppress float drift. This pseudo-observation feedback is the resolution of the mode-3 destabilisation that the earlier $R = 0$ form (18) induced; (18) survives in the code only as the documented, rejected legacy approach.

1.7 Network-producer AR (both mechanisms default off)

The PPP-RTK network producer of the Network mode document is *float-only*. Two integer mechanisms exist in the producer's per-epoch seam sequence, both default-off, and neither changes the emitted satellite-bias product in its shipped state. They are documented here because their algebra is settled, not because they are in use.

Producer AR: a log-only fixability probe. Under `network.ambiguity_resolution.enable` (default `false`) the seam `filter/NetworkProducerAr` reconstructs each station's float ambiguities from the reduced solution (`NetworkEstimator::reconstructStation`), forms an *independent* between-station double-difference basis (a star topology per (constellation, signal), restricted to GPS and Galileo because GLONASS FDMA leaves a per-slot receiver term that survives the difference), builds the exact DD covariance from `NetworkEstimator::jointConditionalCovariance`, rescales it to cycles by (4), and calls `ils::ilsFix`. The result is *logged and discarded*: nothing is re-pinned, nothing is re-solved, and the run is byte-identical whether the block executes or not. The one piece of invariant algebra that makes the probe meaningful is that the between-station DD ambiguity

$$\nabla\Delta N^{i,p_g}_{r_1 r_2,s} \;=\; \Delta N^{i,p_g}_{r_1,s} - \Delta N^{i,p_g}_{r_2,s} \tag{20}$$

is invariant under the producer's choice of phase-bias S-basis: the satellite bias b^i_s cancels in the between-receiver difference exactly as it does in the baseline DD of the double-difference section of the Foundations document, so the probe reports the network's intrinsic integer recoverability rather than a property of the datum it happens to hold. The convergence gauge the probe emits is the median DD formal standard deviation in cycles; it has to fall well below the half-cycle scale before (1) is reachable at all, and it is the quantity to watch rather than the fix count. Two configuration caveats: `ils_max_dim` (default 0 = uncapped) exists because the search does not return on the full-network basis at operational station counts, and caps it to the lowest-formal-σ subset in the manner of Section 1.4; and `wl_cascade` is parsed and echoed in the load-time log but has no consumer anywhere in the tree: it is inert.

SD extra-wide-lane integers: Route B and the A2 datum. The second mechanism (`filter/NetworkSdEwlAr`, gated on `network.sd_ewl.fix_enable`, default `false`) decides an integer for the between-satellite single-differenced extra-wide-lane ambiguity. Its motivation is a rank argument: the float Melbourne–Wübbena row that the enclosing SD-EWL layer folds is a linear combination of observables the epoch's normal equations already contain, so it adds *no new rank*; the only genuinely new content available from that row is integrality. Rounding the SD statistic directly is circular, since it would require the satellite bias difference $b^{i}_{\mathrm{EWL}} - b^{p_g}_{\mathrm{EWL}}$ the network is solving for. Route B breaks the circle by differencing the SD MW statistic *between stations*, which is the same double difference as (20):

$$\boxed{\left(\mathrm{MW}^{i}_{r_1} - \mathrm{MW}^{p_g}_{r_1}\right) - \left(\mathrm{MW}^{i}_{r_2} - \mathrm{MW}^{p_g}_{r_2}\right) \;=\; \lambda_{\mathrm{EWL}}\,\nabla\Delta N^{i,p_g}_{r_1 r_2,\mathrm{EWL}} \;+\; \nabla\Delta\varepsilon^{\mathrm{MW}},} \tag{21}$$

in which the satellite bias pair cancels exactly (same satellite pair on both sides; each station's receiver term already cancelled inside its own single difference), leaving a pure integer against averaged MW noise at the extra-wide-lane wavelength. An integer decided this way uses no a-priori satellite bias and therefore cannot be poisoned by a wrong one, the reason Route B is preferred over rounding against a bias product. The cancellation in (21) carries a precondition that is easy to overlook and is not an implementation detail: b_{EWL} is a per-*signal* combination, so both stations' arcs must sit on the same two tracking channels, or a per-signal bias difference is folded in as though it were an integer. The seam therefore elects a network-common channel pair per satellite by station consensus and admits only stations holding that exact pair.

What (21) yields is a double difference, while the row to be constrained needs a single-differenced integer per (station, satellite). The A2 datum supplies one: within each connected component elect a reference station r_0, define $\Delta n^{i,p_g}_{r_0} := 0$, and propagate the rounded DD fixes outward along a spanning tree, the per-station free integer being absorbed by the link satellite. The asserted quantity is then

$$\Delta n^{i,p_g}_{r} \;=\; \Delta N^{i,p_g}_{r,\mathrm{EWL}} - \Delta N^{i,p_g}_{r_0,\mathrm{EWL}}, \tag{22}$$

the true SD integer minus a constant per (satellite pair, component), which is exactly the classical wide-lane fractional-cycle-bias convention modulo λ_{EWL} [1, 2], and is absorbed harmlessly into the estimated bias the network reports. Two consequences follow from (22) and are load-bearing: the per-station free integer, and hence the spanning tree itself, cancels at the point of use, so tree churn is not a defect to be defended against; and r_0's own integers are absorbed into the constant, making the choice of r_0 a datum decision rather than a measurement.

Acceptance is the scalar case of the battery in Section 1.5: a rounded DD is admitted on its fractional residual, as in (17), together with the $n = 1$ instance of the bootstrap bound (16),

$$P_{s,\mathrm{round}} = \mathrm{erf}\!\left(\frac{1}{2\sqrt{2}\,\sigma}\right), \qquad \sigma \text{ in cycles}, \tag{23}$$

so the same failure-rate semantics apply to a single rounding as to the n-dimensional search. Finally, the mechanism is subject to the standing rule that a stiff, time-linked quantity needs a wired event response: the fixed row writes a λ_{EWL}-quantised value into satellite bias states that run at very small process noise, so an integer that *changes* between epochs would inject a step those states relax out of only over hours. The wired response is withhold-on-disagreement: when this epoch's integer differs from the previous one on a *continuous* arc the fix is withheld for that arc rather than repaired, since on a disagreement epoch neither value is known to be right; an arc that restarted (slip, gap, re-acquisition) replaces the memory without being charged a disagreement, because there the integer legitimately changed. The memory is used only to veto, never to supply a value, which is what keeps this short of a carried zero-process-noise hold; that increment is deliberately not built.

Status. Both mechanisms are default-off and neither has a graded, datum-invariant field result. `network.sd_ewl.fix_enable` is additionally force-disarmed at load unless the float row layer (`network.sd_ewl.enable`) is armed, and that layer is itself inert without `network.phase_leveling` and an MW-class slip detector, since it reads the arc accumulator those build. A reader should assume the emitted network product is float.

1.8 Configuration summary

All tunable thresholds live under the JSON path `estimation.ambiguity_resolution.*`, parsed in `src/core/config/GnssConfigLoad.cpp` with defaults in `include/filter/AmbiguityResolver.hpp`. The invariant estimator theory of Sections 1.1 to 1.6 carries no knobs, since changing it changes the mathematics, not a policy. The two network-producer mechanisms of Section 1.7 are configured elsewhere and separately, under `network.ambiguity_resolution.*` and `network.sd_ewl.*` (parsed in `src/core/config/GnssConfigLoadNetwork.cpp`); both master switches default to `false` and are omitted from the table below.

Symbol	Config key (`...ambiguity_resolution.`)	Default	Role
	`mode`	`0`	0 float / 1 continuous / 2 instantaneous / 3 fix-and-hold
τ_{ratio}	`ratio_threshold`	`3.0`	ratio-test acceptance (15)
$\tau_{\text{ratio}}^{\max}$	`ratio_upper_bound`	`1e4`	degenerate-covariance reject (15)
$P_s^{\min}$	`min_success_rate`	`0.999`	FFR gate (16)
κ_{frac}	`nl_max_per_pair_frac`	`0.15`	per-pair residual cap (17)
θ_{AR}	`elevation_mask`	`20°`	candidate elevation mask
σ^2_{hold}	`hold_variance`	`0.001` m^2	pseudo-obs $\boldsymbol{R}$ (19)
$n^{\text{ep}}_{\min}$	`min_epochs`	`10`	convergence guard
$\sigma^{\max}_{\text{pos}}$	`max_position_sigma_m`	`0.05` m	convergence guard
$\sum \sin^2\theta_{\min}$	`min_arc_info`	`0.0` (off)	per-candidate arc-maturity gate, WL/NL path only (Section 1.4)
	`wl_only`	`false`	MW-WL-only, no NL search
σ^2_{wl}	`wl_only_pseudo_var`	`1e-4` cyc^2	WL-constraint pseudo-obs $\boldsymbol{R}$ (Section 1.3)
	`wl_max_frac`	`0.40`	per-arc WL fractional-residual gate (Section 1.3)
	`enable_partial_ar`	`false`	backward-elimination subset fix (Section 1.4)
	`active_systems`	`[]` (all)	constellation subset (Section 1.4)
	`state_ambiguity_basis` †	`per_band`	`per_band` vs `nl_wl` dispatch

† Path is `estimation.state_ambiguity_basis`, one level above the `ambiguity_resolution` block.

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